

ENHANCING EFFICIENCY OF POWER AND ENERGY SUPPLY CHAIN USING SCADA

An Integrated National Energy Management Dashboard (INEMD) Approach, applied to the 4 October 2022 National Blackout. By 2023, Bangladesh's NLDC had a modern SCADA/EMS for power-side monitoring, and GTCL's newer transmission projects included dedicated SCADA/RTU at new Town Border Stations. Yet no real-time, API-mediated bridge links power-side SCADA (NLDC-Ghorashal, BPDB/PGCB) with gas-side SCADA (TGTDCL/GTCL); each utility grew more instrumented internally, but the cross-sector seam remained unaddressed.

This research asks whether an INEMD a real-time coordination layer across gas, power, and petroleum supply chains could have reduced the severity of the 4 October 2022 blackout, and whether it is economically justified as national infrastructure. Two anchor points are evaluated: an hourly linear program optimizing cost-minimal gas allocation from Narsingdi TBS across four downstream branches, penalizing shortfalls and backup-fuel use; and a Monte Carlo restoration-time model of SCADA data sharing between NLDC-Ghorashal and Narsingdi TBS, translating gains into Energy-Not-Served and Value of-Lost-Load savings. A Python pipeline combining `scipy.optimize.linprog`, 10,000-draw simulation, and a 10-year discounted cash-flow appraisal underpins the analysis.

4.5% lower daily system cost under a REAL, reported crisis-day severity; Ghorashal's delivered flow raised from the real 58.6 to ~80.6 MMCFD/day; 126,621 L/day diesel-equivalent avoided. 82.4% modelled reduction in mean restoration time; the baseline model independently validates against PGCB's real switching log/load curve to within 5.6% (ENS) and 15.5%/1.0% (mean/P90 restoration time). US\$11.95M avoided per event (~BDT 123.1 crore), 83.4% reduction, now anchored to a REAL (not modelled) baseline capital loss of US\$14.33M.

A Hybrid Real-Time Early-Warning Framework for Blackout Prevention

Fusing Dragonfly-Optimized Support Vector Regression (DFO-SVR) Voltage Stability Forecasting with Multilevel Flow Modeling (MFM) Root-Cause Analysis – Application to the August 14, 2003 Northeast Blackout Using Reconstructed PMU Data and Automated SCADA-to-EMS Integration. Blackout is a phenomenon that occurs not because of a single cause, but due to multiple cascading disturbances. Most of the large blackouts have been preceded by a short time period, from several minutes up to a few hours, where smaller problems occurred and fault indications and alarms started to pour in to the control room. The operators quickly became overloaded with information and were unable to understand the fault state and to plan and execute actions to save the grid. In recent years, power systems have operated closer to their stability limits because new power system infrastructure construction has lagged behind the steady increment of electricity demand and the increasing practice of long distance bulk power transmission.

I present a hybrid online platform that merges a Dragonfly Optimized Support Vector Regression (DFO-SVR) model for second-by-second Voltage Stability Index (VSI) forecasting and Multilevel Flow Modelling (MFM) root-cause analysis for instantaneous alarm condensation.

Applied to the August 14, 2003 Ohio blackout, the system predicts VSI collapse 25 min ahead ($RMSE < 0.03$) while isolating the Sammis-Star 345 kV trip as the primary root cause within less than 1s. **It can reconstruct PMU-like voltage measurements from SCADA logs.** That means The architecture is plug-and-play, runs on existing EMS infrastructure (traditional grid) and scales to ISO-level grids (smart grid).